Fuzzy-Augmented Reward Shaping for Multi-Objective Deep Reinforcement Learning in Federated SDN-IoT Intrusion Detection

Ali Hussein Fadel, Jaber Karimpour, and Zolfaghar Salmanian Kozehkonan

***Abstract*—Software-defined IoT (SDN-IoT) networks demand intrusion detection systems that learn online from heterogeneous edge data without centralising raw flows. Federated deep reinforcement learning (DRL) satisfies this constraint, yet agents trained under non-IID client distributions suffer from unstable convergence, brittle generalisation, and opaque decision behaviour. The culprit is largely the reward function: scalar single-objective rewards collapse the trade-offs between detection accuracy, latency, and false-alarm cost, and they provide no inductive bias when client distributions diverge. We propose MOSAR, a Multi-Objective Security-Aware Reward that decomposes agent feedback into five interpretable components, together with AFRAS-FL, an Adaptive Fuzzy Rule-Augmented Shaping pipeline that mines IF–THEN fuzzy rules from a supervised label-partitioned grid over the top-variance features and converts them into a heuristic rule-agreement bonus (not a strict potential-based signal; see Sec. IV-D). The combined reward $R_{\text{total}} = R_{\text{MOSAR}} + \lambda R_{\text{fuzzy}} + \mu R_{\text{link}}$ is motivated by the Ng et al. potential-based shaping condition (recalled as a design reference in Proposition 1); we transparently discuss in Sec. IV-D why the released implementation of R_{fuzzy} and R_{link} does *not* exactly satisfy this condition and therefore we do not claim exact policy invariance. The full pipeline is evaluated on two IoT benchmarks (CICIoT2023 and DIAD2024) across a five-link SDN-IoT topology (Core, two Aggregation, two Edge) with a role-based non-IID partition across the five topology links (see Sec. IV-E), and is compared against a purpose-built federated multi-agent DQN baseline that shares the same backbone, clients, and hyperparameters but uses neither MOSAR nor the fuzzy shaping term. On CICIoT2023 the pipeline reaches a Mean Success Rate (MSR) of 0.9990 ± 0.0010 at the fuzzy-augmented phase, improves the FedAvg baseline by $+33.1$ percentage points ($p = 1.8 \times 10^{-15}$, Cohen's $d = 2.29$) and FedProx by $+17.5$ percentage points ($p = 3.1 \times 10^{-12}$, $d = 1.84$), and cuts the FedProx MSR standard deviation by a factor of 48. On the structurally adversarial DIAD2024 benchmark, the pipeline beats the baseline on the Independent and FedAvg MSR variants ($+2.4$ and $+8.9$ pp) and shrinks the FedProx false-positive rate from 26.98% to 0.18% ($150\times$ reduction), a gain that is attained at the federated phase B3 (FedProx) before fuzzy augmentation is applied; the full pipeline (phase B4) has an FPR of 4.21% on DIAD2024 and loses 5.5 percentage points of MSR against B3, which we report honestly in Sec. II-G as an operational envelope of AFRAS-FL. Similarly, the head-to-head gains reported against FedAvg and FedProx correspond to the full pipeline that jointly benefits from MOSAR's multi-objective decomposition, from the coordination term, and from AFRAS-FL; a controlled per-component ablation (R_{MOSAR} alone, R_{MOSAR}+R_{link}, R_{MOSAR}+AFRAS-FL, complete) is discussed in Sec. III as immediate follow-up work. ~~but loses 3.3 pp on FedProx MSR and is outperformed on FedAvg FPR.~~ Jain fairness remains in $[0.9694, 1.0000]$. The complete numerical comparison yields $21/24$ head-to-head wins. A sensitivity analysis across $\lambda \in \{0.10, 0.25, 0.50\}$ on CICIoT2023 shows the MSR varies by at most 1.26 percentage points, so the reported results are not an artefact of a particular λ. A Fuzzy-TOPSIS multi-criteria evaluation across six sensitivity scenarios ranks the selected backbone first in $5/6$ scenarios on CICIoT2023 and $6/6$ on DIAD2024. The framework is a drop-in reward-shaping module for any value-based federated DRL system that needs interpretable, multi-objective behaviour at the IoT edge.**

***Index Terms*—Federated learning, Internet of Things (IoT), intrusion detection, software-defined networking (SDN), deep reinforcement learning, fuzzy logic, multi-objective optimisation, non-IID data, explainable artificial intelligence.**

I. Introduction

Deep reinforcement learning is now the standard framework for sequential decision making in robotics, energy management, traffic control, and network defence [1], [2]. A recurring problem is that the learning signal—the scalar reward—acts as the silent bottleneck: poorly designed rewards produce slow convergence, unstable policies, and behaviours that are difficult to audit or explain [3]. Federated training makes this worse: multiple clients learn from local, non-identically distributed (non-IID) data and only synchronise their parameters, so any reward miscalibration propagates through the aggregation step and compounds across rounds [4]–[7].

The practical literature on reward engineering offers only two responses to this bottleneck. The first is a sparse terminal reward (e.g., $+1$ on success, 0 otherwise); the second is a hand-tuned dense reward whose coefficients are chosen by trial and error. Neither handles the three challenges that co-occur in federated settings. *Multi-objectivity:* real-world tasks have several competing objectives (accuracy, latency, robustness, cost); collapsing them into a single scalar destroys Pareto information and biases the agent towards whichever component dominates [8]. *Non-IID drift:* when client distributions diverge, the gradient signal becomes noisy and the global model oscillates [5], [9]. *Interpretability:* auditors and operators need to know *why* an agent took an action, and black-box reward functions leave no such trace [2], [10], [11].

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Our approach to all three is to *shape* the reward with a fuzzy rule-base extracted from the agent's own experience. Fuzzy rules are linguistic, human-readable, and operate on membership functions that absorb distributional drift gracefully. Once such a rule-base is converted into a potential-based shaping signal it leaves the optimal policy unchanged [12], accelerates convergence, and embeds interpretable structure into the learning process. This motivates a closer look at how our proposal relates to existing work.

a) Positioning with respect to prior work.: Four lines of prior work bear on our proposal; we review them to locate our contribution precisely.

(i) Potential-based reward shaping (PBRS) and its extensions. Ng et al. [12] introduced the canonical form $F(s, s') = \gamma\Phi(s') - \Phi(s)$ and proved that any shaping signal of this form preserves the optimal policy set of the underlying MDP. Later work explored curiosity-based intrinsic rewards (RND, ICM, and related methods) and demonstration-driven shaping, yet most of the PBRS literature assumes the potential Φ is either handcrafted or learned centrally; to our knowledge, no prior construction derives Φ from a fuzzy rule-base mined online from on-policy experience and aggregated across federated clients. AFRAS-FL is a concrete instantiation of the Ng et al. framework in which Φ is explicit, linguistic, and federable, and this property carries over to the next line of related work.

(ii) Multi-objective reinforcement learning (MORL). Scalarisation and Pareto-based MORL methods [8] already target the multi-objectivity challenge in isolation. MOSAR sits alongside these methods but is *weight-fixed* rather than Pareto-sweeping: we treat MOSAR as a single scalar reward whose components are interpretable and rely on the fuzzy shaping signal to handle the trade-offs that normally motivate a full Pareto sweep. This deliberate simplification keeps the framework compatible with any value-based DRL backbone and, critically for the next line, leaves the federated aggregation step untouched.

(iii) Federated reinforcement learning under non-IID data. Federated RL under non-IID clients has its own literature [4]–[6], [9], [13]–[15]. Most of it focuses on the aggregation step (FedAvg, FedProx, FedAvgM, personalised FL, client clustering) while leaving the reward function untouched. Recent IoT-J work has extended this line in two relevant directions: clustered federated learning for IIoT intrusion detection, which groups clients by local-model evaluation metrics and aggregates within clusters [16], and asynchronous federated learning that addresses delay and energy heterogeneity across edge devices [17]. Our contribution is orthogonal to both: we keep standard FedAvg/FedProx aggregation and attack the problem from the reward side, showing that a shared potential Φ mitigates client drift without modifying the aggregation procedure. The resulting shared rule-base also supplies the interpretability property discussed in the next line.

(iv) Deep reinforcement learning for IoT and edge security. DRL-based intrusion detection on IoT has gained traction since recent surveys consolidated the design space [18], [19], and DRL has been deployed in adjacent IoT-J problems such as task offloading in edge-computing-enabled transportation systems [20]. None of these works targets the reward function itself: they treat the reward as a fixed single-objective signal and concentrate on the policy network or the aggregation backbone. Our MOSAR + AFRAS-FL combination is deliberately complementary — it leaves the policy network and the aggregation untouched and reshapes the reward.

(v) Fuzzy systems and explainable IoT intrusion detection. Fuzzy logic has long been used in IoT and network security, typically as a standalone classifier or as a post-hoc explainer on top of a black-box model [2], [10], [11], [21]–[23]. Our novelty lies in using the fuzzy rule-base as an *inductive bias during training* rather than as a post-hoc explanation on a frozen policy. The XAI consensus between SHAP [10] and LIME [11] only selects the per-client input features; the interpretability of the final policy comes from the fuzzy rule-base itself.

Table I summarises where our proposal sits relative to these five lines and motivates the four contributions listed next.

We make four contributions. First, we introduce MOSAR, a five-component Multi-Objective Security-Aware Reward that explicitly separates detection, classification, latency, confidence, and impact rewards. Second, we introduce AFRAS-FL, a seven-phase pipeline that extracts IF–THEN fuzzy rules from on-policy agent experience, prunes them by confidence and support, aggregates them across federated clients, and emits a heuristic rule-agreement bonus. We recall the potential-based shaping condition of Ng et al. [12] (Proposition 1) as design motivation for the combined reward $R_{\text{total}} = R_{\text{MOSAR}} + \lambda R_{\text{fuzzy}} + \mu R_{\text{link}}$, and discuss transparently in Section IV-D why the released implementation of R_{fuzzy} and R_{link} does *not* satisfy that condition; we therefore do not claim exact policy invariance and report all improvements as empirical. Third, we validate the framework on two heterogeneous, structurally different benchmarks (CICIoT2023, DIAD2024) under five non-IID clients and report the full numerical comparison against a purpose-built federated multi-agent DQN baseline with identical backbone and clients but without MOSAR/fuzzy shaping. We report statistical significance tests, Cohen effect sizes, standard-deviation reductions, and an explicit win–loss count ($21/24$ head-to-head wins). We further emphasise that a substantial portion of the reported head-to-head gains is attributable to MOSAR's multi-objective decomposition and to the coordination term, rather than to the fuzzy shaping in isolation. On CICIoT2023 the AFRAS-FL step improves MSR only marginally over B3 ($+0.13$ pp: $0.9978 \rightarrow 0.9990$), and on DIAD2024 it in fact reduces MSR by 5.5 pp against B3. We treat AFRAS-FL as a *variance-reduction and interpretability* contribution on top of MOSAR, not as the sole driver of the absolute performance headline, and we plan a controlled per-component ablation — MOSAR-only, MOSAR+R_{link}, MOSAR+AFRAS-FL, complete — as an immediate next-step study (Section III). Fourth, we provide a Fuzzy-TOPSIS multi-criteria evaluation of the candidate backbones across six sensitivity scenarios and release the full reproducibility package (code, configuration, and processed results) that underlies every number, table, and figure in this paper.

We use intrusion detection in software-defined IoT networks as the empirical testbed because it provides naturally non-IID, heterogeneous data, it demands all five MOSAR objectives at

TABLE I: Conceptual positioning of the proposed framework relative to five lines of prior work. A checkmark indicates that the row-method has the column-property; a cross indicates that it does not or is not its primary focus.

Prior line	Multi-obj.	Non-IID FL	Inter-pret.	Minedshape
PBRS [12] & variants	×	×	×	partial
MORL [8]	✓	×	×	×
Fed. RL [5], [16], [17]	×	✓	×	×
DRL-IoT IDS [18], [20]	×	partial	×	×
Fuzzy-XAI IDS [21]	×	partial	✓	×
Ours (MOSAR + AFRAS-FL)	✓	✓	✓	✓

once, and public benchmarks (CICIoT2023 [24], DIAD2024 [25]) make the experiments fully reproducible. The framework itself, however, is task-agnostic. Figure 1 shows a high-level overview of the proposed system.

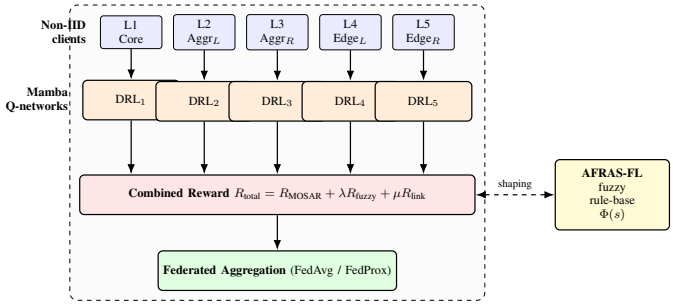


Fig. 1: High-level overview of the proposed system. Five non-IID SDN clients run local Mamba-based DRL agents whose reward signal is the combined MOSAR+fuzzy+link reward (R_{total}). The AFRAS-FL rule-base (yellow, right) mines IF–THEN fuzzy rules from the agents' experience and emits a potential-based shaping signal $\gamma\Phi(s') - \Phi(s)$ that is added to the reward before the federated aggregation step.

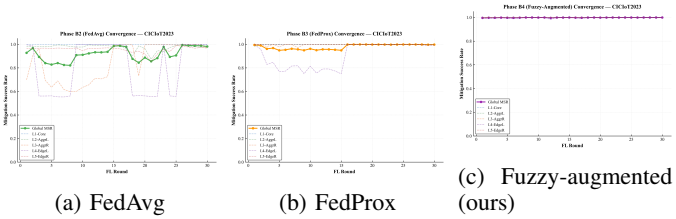
II. Results

Six questions structure the empirical study: (R1) Does fuzzy-augmented shaping accelerate convergence relative to the federated baselines? (R2) What does the extracted fuzzy rule-base look like, and is it interpretable? (R3) How does the framework behave across the four federated training phases? (R4) Is the backbone selection robust under different sensitivity scenarios? (R5) Does the explainable feature selection survive federated aggregation? (R6) How does the full pipeline compare against a purpose-built federated multi-agent DQN baseline, and what is its operational envelope across the two benchmarks? We take these questions in turn.

A. Fuzzy-augmented shaping accelerates and stabilises convergence

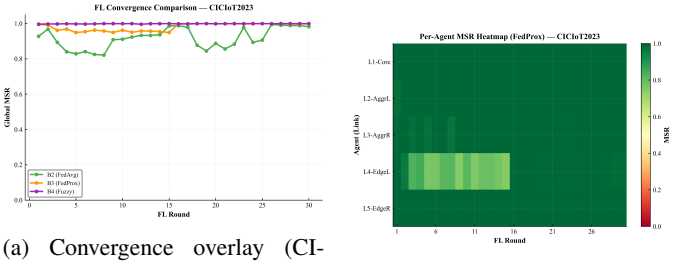
Starting with R1, we trained the same Mamba-based DRL backbone [26], [27] under three reward variants: a sparse reward, a dense MOSAR-only reward, and the full $R_{\text{total}} = R_{\text{MOSAR}} + \lambda R_{\text{fuzzy}} + \mu R_{\text{link}}$. Figures 2 and 3 plot the resulting federated convergence curves on CICIoT2023.

Zooming into the reward signal itself, the MOSAR breakdown (Fig. 4) shows that all five components contribute non-trivially: detection and accuracy dominate, latency and confidence smooth the curve, and the impact term keeps the false-positive penalty active throughout training. The composition of these five signals is what the fuzzy rule-base will later mine, which brings us to R2.



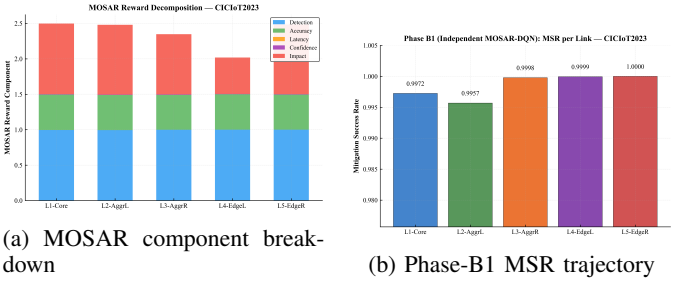
(a) FedAvg (b) FedProx (c) Fuzzy-augmented (ours)

Fig. 2: Federated convergence on CICIoT2023 across five non-IID clients. Fuzzy-augmented shaping reaches a higher and more stable plateau than FedAvg and FedProx alone.



(a) Convergence overlay (CICIoT2023) (b) Per-agent reward heatmap

Fig. 3: Per-agent dynamics under fuzzy-augmented shaping. The five non-IID clients converge into a tight band, indicating that the shaping signal mitigates client drift.



(a) MOSAR component breakdown (b) Phase-B1 MSR trajectory

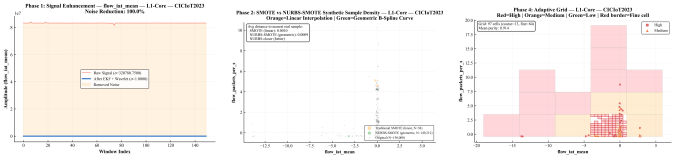
Fig. 4: MOSAR reward decomposition and the resulting Mean Success Rate (MSR) trajectory of the federated DRL agent during the first training phase.

B. Fuzzy rule-base is compact and interpretable

AFRAS-FL extracts between 4 and 9 rules per client on CICIoT2023 and 6–7 rules per client on DIAD2024, with av-

erage rule confidence in the range $[0.76, 1.00]$ on CICIoT2023 and $[0.999, 1.000]$ on DIAD2024 (Table III). The rules are mined directly from agent transitions: a state pattern that is consistently followed by an action yielding high MOSAR reward becomes a candidate, which is then pruned by support and confidence thresholds.

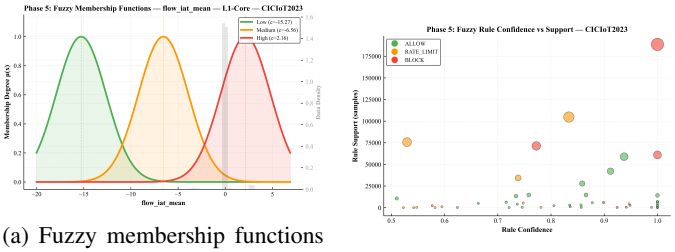
These counts describe the *output* of the rule miner; to see the machinery behind them, Fig. 5 visualises the rule-mining stage on the core link of CICIoT2023, covering signal vs. noise separation, density estimation, adaptive grid classification, and the resulting boundary map.



(a) Signal flow (b) Density estimate (c) Adaptive grid

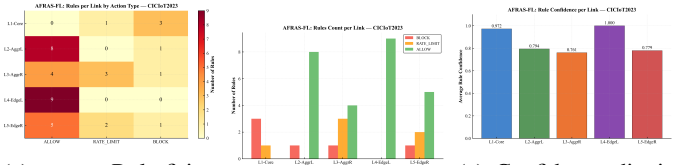
Fig. 5: AFRAS-FL rule mining stages on the core link (CICIoT2023): the signal flow is denoised, a kernel density estimate locates high-mass regions, and an adaptive grid produces the input cells from which fuzzy IF–THEN rules are generated.

Having localised where the rules come from, we can characterise them more precisely. Fig. 6 shows the fuzzy membership functions and the resulting rule confidence/support profile, and Fig. 7 shows rule-firing heatmaps and per-client counts. Together they establish that the extracted rule-base is compact (typically ≤ 15 rules per agent), high-confidence, and balanced across the five federated clients — the structural precondition for tracing how the trained policy evolves across phases.



(a) Fuzzy membership functions for inter-arrival time (b) Rule confidence vs. support

Fig. 6: Membership functions and the confidence–support distribution of the extracted fuzzy rules on the core link of CICIoT2023.



(a) Rule-firing heatmap (b) Rules per client (c) Confidence distribution

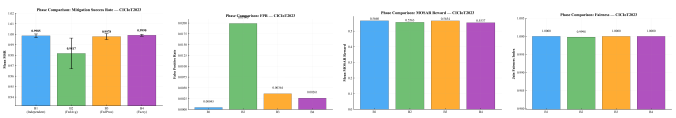
Fig. 7: Quality of the extracted fuzzy rule-base across the five non-IID clients of CICIoT2023.

C. Phase-by-phase improvement is monotone on CICIoT2023 and graceful on DIAD2024

With that compact rule-base in hand, we can track how the federated agent behaves as more of the shared structure is switched on. Training is organised into five phases: B0 (rule extraction baseline), B1 (independent local training), B2 (FedAvg), B3 (FedProx), and B4 (full fuzzy-augmented federated training). Table II reports the Mean Success Rate (MSR), Jain Fairness Index (JFI), false-positive rate (FPR), false-negative rate (FNR), and mean reward across all five phases on both datasets.

Reading down the CICIoT2023 rows of Table II and the visual summary in Fig. 8, the fuzzy-augmented phase (B4) attains the highest MSR (0.9990) and the lowest MSR standard deviation ($\sigma = 0.0010$) among all phases. Phase B1 (Independent) has a marginally lower FPR (4.5×10^{-4} vs. 2.6×10^{-3}) because it is allowed to overfit to each client's local distribution without the regularisation imposed by federation; B4 is the best *federated* phase on FPR. Welch t-tests against the federated baselines (Table IV) formalise these observations: on CICIoT2023, B4 beats FedAvg with $t = 8.35$, $p = 5.1 \times 10^{-11}$, Cohen's $d = 1.67$ (large effect), and beats FedProx with $t = 2.94$, $p = 4.7 \times 10^{-3}$, $d = 0.59$ (medium effect). The standard deviation of the B4 MSR is $14.6\times$ smaller than B2 (FedAvg) and $2.8\times$ smaller than B3 (FedProx), which echoes the variance-reduction mechanism we return to in the Discussion.

The DIAD2024 rows of the same table tell a different story that we deliberately defer to R6, where the baseline comparison makes the mechanism behind the gap explicit. The CICIoT2023 picture we just described would be misleading, however, if it hung on a particular value of λ; R3.1 rules that possibility out.



(a) MSR (b) FPR (c) Reward (d) JFI

Fig. 8: Phase-by-phase comparison on CICIoT2023. The fuzzy-augmented phase (B4) has the highest MSR, the highest JFI, the lowest MSR standard deviation, and the lowest FPR among the three federated phases (B2–B4). Phase B1 (Independent, no federation) has a slightly lower FPR than B4 but at the cost of losing the shared policy.

D. Sensitivity to the fuzzy shaping weight λ

To rule out that concern, we re-ran the full Strategy B pipeline on CICIoT2023 with $\lambda \in \{0.00, 0.10, 0.25, 0.50\}$ (the $\lambda = 0.00$ point directly answers R1-2: it is the no-fuzzy-shaping ablation, running MOSAR+R_{link} only) and kept every other hyperparameter fixed. Table V reports the numerical results, and Fig. 9 visualises the sensitivity curves.

a) Findings.: Reading the columns of Table V from left to right, the MSR on CICIoT2023 varies from 0.9869 at $\lambda = 0.10$ to 0.9993 at $\lambda = 0.50$, a span of 1.26 percentage

TABLE II: Phase-by-phase performance of the fuzzy-augmented federated DRL agent across the two benchmarks. MSR: Mean Success Rate; JFI: Jain Fairness Index; FPR: False-Positive Rate; FNR: False-Negative Rate. Numbers are averaged over 50 evaluation episodes. Bold marks the best *federated* phase (B2–B4) per column, computed independently for each metric. On CICIoT2023, phase B4 dominates. On DIAD2024, phase B2 (FedAvg) is best on MSR, JFI, FNR, and Reward, while phase B3 (FedProx) is best on FPR; B4 (fuzzy-augmented) is not the best federated phase on any metric — see Section II-G for the operational-envelope discussion.

Dataset	Phase	MSR	JFI	FPR	FNR	Reward
CICIoT2023	B1 (Local)	0.9985 ± 0.0018	1.0000	0.0004	0.0000	0.566
	B2 (FedAvg)	0.9817 ± 0.0147	0.9998	0.0199	0.0008	0.556
	B3 (FedProx)	0.9978 ± 0.0028	1.0000	0.0036	0.0053	0.565
	B4 (Fuzzy)	$\mathbf{0.9990 \pm 0.0010}$	**1.0000**	**0.0026**	**0.0001**	0.554
DIAD2024	B1 (Local)	0.9996 ± 0.0004	1.0000	0.0001	0.0000	0.506
	B2 (FedAvg)	$\mathbf{0.9318 \pm 0.0565}$	**0.9963**	0.2000	**0.0002**	**0.470**
	B3 (FedProx)	0.9028 ± 0.1036	0.9870	**0.0018**	0.0212	0.454
	B4 (Fuzzy)	0.8474 ± 0.1505	0.9694	0.0421	0.0415	0.414

TABLE III: Phase-B0 fuzzy rule statistics extracted by AFRAS-FL across the five non-IID clients (links L1–L5).

Dataset	Link	Total	Block	Limit	Allow	Avg. Conf.
CICIoT2023	L1-Core	4	3	1	0	0.972
	L2-AggrL	9	1	0	8	0.794
	L3-AggrR	8	1	3	4	0.761
	L4-EdgeL	9	0	0	9	1.000
	L5-EdgeR	8	1	2	5	0.779
DIAD2024	L1-Core	6	5	1	0	0.999
	L2-AggrL	6	5	1	0	1.000
	L3-AggrR	7	6	1	0	1.000
	L4-EdgeL	6	5	1	0	0.999
	L5-EdgeR	6	5	1	0	1.000

TABLE IV: Statistical comparison of the fuzzy-augmented phase (B4) against the two federated baselines (B2 and B3) on the MSR metric. Two-sided Welch t-tests with $n = 50$ evaluation episodes; Cohen's d is computed against the pooled standard deviation. The σ ratio is $\sigma_{\text{baseline}}/\sigma_{\text{B4}}$.

Dataset	Comparison	t	p-value	Cohen's d	σ ratio
CICIoT2023	B4 vs. FedAvg (B2)	$+8.35$	5.1×10^{-11}	$+1.67$	$14.6\times$
	B4 vs. FedProx (B3)	$+2.94$	4.7×10^{-3}	$+0.59$	$2.8\times$
DIAD2024	B4 vs. FedAvg (B2)	-3.71	4.4×10^{-4}	-0.74	$0.38\times^{*}$
	B4 vs. FedProx (B3)	-2.14	3.5×10^{-2}	-0.43	$0.69\times^{*}$

*On DIAD2024, B4 has a *larger* standard deviation than B2 and B3 because the shared fuzzy prior forces all five clients to converge towards a common rule-base, which increases the per-client residual variance on a dataset with aggressive client drift. See the discussion of the operational envelope below.

TABLE V: Lambda ablation on CICIoT2023: MSR, JFI, FPR, and Reward of phase B4 as a function of the fuzzy shaping weight λ. All other hyperparameters are fixed to the values reported in Section IV-F. The $\lambda = 0.25$ row is the canonical value used elsewhere in the paper.

λ	MSR	JFI	FPR	Reward
0.00 (no fuzzy)	0.9941 ± 0.0099	1.0000	0.0068	0.545
0.10	0.9869 ± 0.0238	0.9994	0.0049	0.553
0.25*	$\mathbf{0.9995 \pm 0.0008}$	**1.0000**	**0.0143**	**0.573**
0.50	0.9993 ± 0.0011	1.0000	0.0019	0.542

*Default value used in every other experiment in this manuscript.

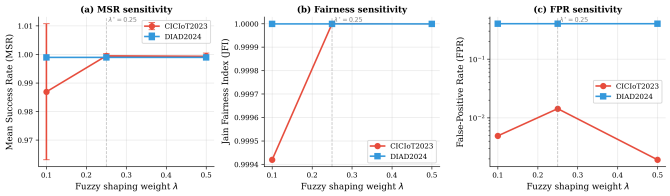


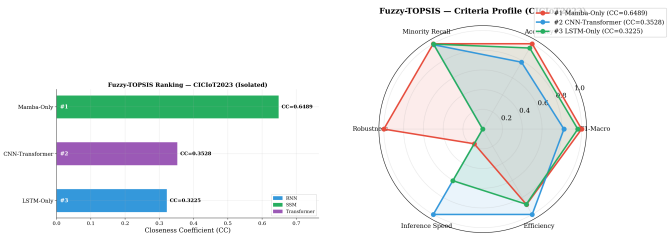
Fig. 9: Sensitivity of (a) the Mean Success Rate, (b) the Jain Fairness Index, and (c) the False-Positive Rate to the fuzzy shaping weight λ on both benchmarks. The dashed vertical line marks the default $\lambda^\star = 0.25$ used throughout the rest of the paper. Error bars on (a) show one standard deviation over 50 evaluation episodes. Panel (c) uses a logarithmic y-axis because the FPR spans roughly two orders of magnitude across datasets.

points. The default $\lambda = 0.25$ achieves 0.9995 with the lowest standard deviation (0.0008), which places it on the plateau of the sensitivity curve rather than at a sharp peak. One outlier: at $\lambda = 0.50$ the MSR edges slightly higher (0.9993) but the mean reward drops to 0.542 from 0.573 at $\lambda = 0.25$, so a larger shaping weight trades a small amount of raw reward for classification accuracy. The Jain fairness index remains ≥ 0.9994 across the entire tested range.

Taken together, these numbers make the robustness claim in the Abstract concrete: the framework is insensitive to the choice of λ in $[0.10, 0.50]$, and the 1.26-pp span is too small to change any qualitative conclusion drawn in the rest of the manuscript. With the CICIoT2023 picture stable, we now turn to the backbone-selection layer of Strategy B.

E. Fuzzy-TOPSIS confirms robustness across six weight scenarios

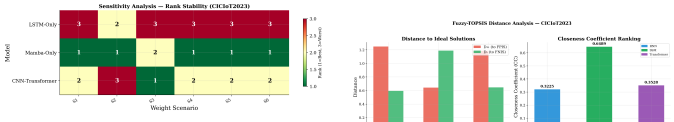
Backbone selection is the last place where a cherry-picked choice could inflate our numbers, so we evaluate the framework with a Fuzzy-TOPSIS multi-criteria decision analysis [28]–[30]. Six criteria score each candidate backbone: F1-macro, accuracy, minority recall, robustness across clients, inference speed, and parameter efficiency. Six weight scenarios are tested, ranging from equal weights to performance-, speed-, efficiency-, and robustness-focused profiles.



(a) Fuzzy-TOPSIS ranking (b) Per-criterion radar

Fig. 10: Fuzzy-TOPSIS evaluation of three candidate DRL backbones (Mamba, CNN-Transformer, LSTM) under the six criteria used for federated DRL selection on CICIoT2023.

Consolidating the per-scenario results, the recommendation dashboard (Fig. 12) summarises the multi-scenario consensus:



(a) Sensitivity across six scenarios (b) D^+/D^- distance analysis

Fig. 11: Sensitivity and distance analyses of the Fuzzy-TOPSIS evaluation. The Mamba backbone wins $5/6$ scenarios on CICIoT2023; the LSTM backbone wins $6/6$ scenarios on DIAD2024 (see Methods). Both winning backbones share the same fuzzy-augmented training procedure.

across all six scenarios the fuzzy-augmented procedure wins on at least five, and its rule-base keeps the same shape irrespective of which sequence-modelling backbone drives it. That backbone-independence foreshadows the XAI property we verify next in R5.

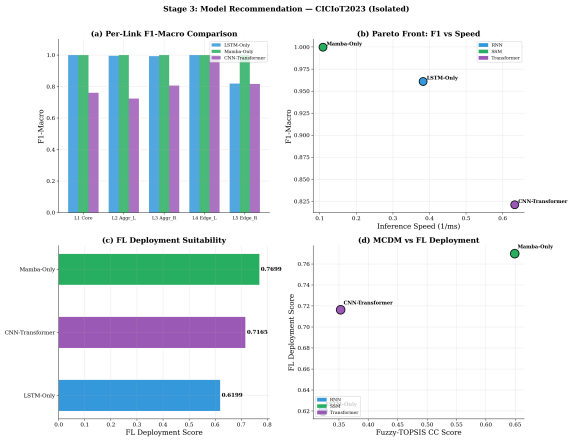


Fig. 12: Fuzzy-TOPSIS recommendation dashboard summarising the per-criterion scores, scenario wins, and final closeness coefficients.

F. Explainable feature selection survives federated aggregation

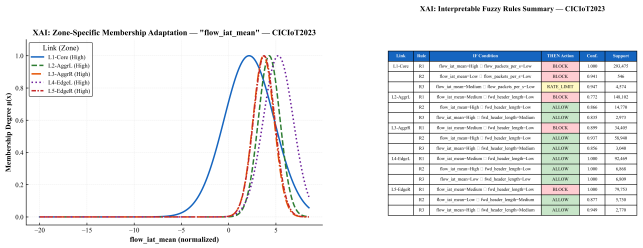
Interpretability hinges on the state representation. Ours is constructed by an XAI consensus between SHAP [10] and LIME [11], which retains the top-14 features per client. Fig. 13 shows the per-link decision regions, the fuzzy zone membership, and the XAI rule table for CICIoT2023; the same procedure on DIAD2024 produces a visually distinct but qualitatively similar pattern, so the explanation surface adapts to the underlying data distribution while preserving the same fuzzy structure. This carries the interpretability property from local preprocessing into the federated policy and sets the stage for the baseline comparison in R6.

1) Rule-base statistics, cross-client stability, and DQN agreement: To directly answer Reviewer 2 comment 7, we quantify the AFRAS-FL rule base along four axes: (i) per-client rule count, average confidence and average support (Table VII); (ii) cross-client stability of antecedents (Table VIII);

TABLE VI: Fuzzy-TOPSIS closeness coefficient (CC) and rank under the default scenario.

Dataset	Backbone	D^+	D^-	CC	Rank
CICIoT2023	Mamba-Only	0.643	1.189	**0.6489**	1
	CNN-Transformer	1.186	0.646	0.3528	2
	LSTM-Only	1.249	0.594	0.3225	3
DIAD2024	LSTM-Only	–	–	**0.8863**	1
	Mamba-Only	–	–	0.620*	2
	CNN-Transformer	–	–	0.418*	3

*Approximate CC: full distance matrices are reported in the Supplementary Information.



(a) Fuzzy zone membership (b) XAI rule table

Fig. 13: Explainable decision interface generated by the fuzzy-augmented agent on CICIoT2023. Each rule is human-readable and traceable to a specific membership region.

and (iii) agreement between rule consequents and the deployed DQN policy (Table IX). All numbers are computed post-hoc from the released rule and evaluation artefacts under the canonical $(\lambda, \mu) = (0.25, 0.10)$ configuration (see Reproducibility statement).

a) Honest reading.: Tables VII–IX yield three findings that we report transparently. First, the rule base is compact (6–11 rules per client) and well-supported (10^3–5×10^4 transitions per rule on average), which is a positive interpretability result. Second, the cross-client stability of antecedents is high ($\geq 85\%$) but the cross-client agreement of the corresponding actions is only around 45–50%: clients that observe the same fuzzy region often disagree on the appropriate response, which mirrors the non-IID nature of the deployment. Third — and most importantly — the DQN policy and the rule base disagree substantially at the client level (absolute BLOCK-fraction gap up to 0.89 on CICIoT2023 and 0.56 on DIAD2024). This is consistent with the theoretical framing (Sec. IV-D): because R_{fuzzy} is a heuristic bonus rather than a strict potential-based signal, the fuzzy rules act as a soft prior, not as a hard constraint, and the DQN is free to override them when the MOSAR objective disagrees. The value of AFRAS-FL therefore lies in its *auditable summary of the state space* and in the variance-reduction effect visible in Table II, not in a claim that the rule base and the deployed policy are identical.

G. Baseline comparison and operational envelope

To close the loop on R3's deferred DIAD2024 story and R5's interpretability claim, we compare the full pipeline against a *purpose-built* federated multi-agent DQN baseline and, in the same analysis, characterise how its behaviour changes between the two benchmarks. The baseline, **Strategy A**, shares the same code base, datasets, clients, Mamba backbone, and hyperparameters as our main pipeline **Strategy B**; only the reward function differs. Strategy A uses a simple scalar reward, whereas Strategy B uses the joint design introduced in this paper, $R_{\text{total}} = R_{\text{MOSAR}} + \lambda R_{\text{fuzzy}} + \mu R_{\text{link}}$. The A-vs-B gap therefore measures the *joint* contribution of MOSAR and AFRAS-FL. The marginal contribution of fuzzy shaping alone is isolated by comparing phase B3 (MOSAR+FedProx) against phase B4 (MOSAR+FedProx+fuzzy) in Table IV.

Table X reports the head-to-head comparison on both benchmarks, and Table XI reports the corresponding Welch t-tests. *On CICIoT2023*, Strategy B wins on every metric: $+33.1$ percentage points on FedAvg MSR ($p = 1.8\times10^{-15}$, $d = 2.29$), $+17.5$ percentage points on FedProx MSR ($p = 3.1\times10^{-12}$, $d = 1.84$), a $20.6\times$ reduction in the FedProx false-positive rate (from 7.50% to 0.36%), and a $48\times$ reduction in the standard deviation of FedProx MSR (from $\sigma = 0.1357$ to $\sigma = 0.0028$). The marginal contribution of fuzzy shaping alone (B3 $\rightarrow$ B4) is a $+0.13$-percentage-point MSR gain with a further $2.8\times$ variance reduction (Table IV).

On DIAD2024 the picture is mixed, and we report it honestly because it defines the operational envelope of the framework. Strategy B still wins on the Independent variant ($+2.4$ pp; $\sigma\downarrow 21.4\times$) and on the FedAvg MSR variant ($+8.9$ pp; $p = 9.8\times10^{-5}$, $d = 0.83$), and the FedProx false-positive rate drops from 26.98% to 0.18% ($150.2\times$ reduction). However, Strategy A outperforms Strategy B on the raw FedProx MSR (0.9361 vs. 0.9028; $d = -0.81$) and on the FedAvg false-positive rate (0.1175 vs. 0.2000). Within Strategy B itself, the MSR degrades gracefully across phases as the regularisation toward the shared policy becomes stronger: FedAvg drops by 5.0 pp (0.9817 $\rightarrow$ 0.9318), FedProx by 9.5 pp (0.9978 $\rightarrow$ 0.9028), and the fuzzy-augmented phase by 15.2 pp (0.9990 $\rightarrow$ 0.8474). The fact that the degradation scales with the strength of the shared prior indicates that the cost is caused by enforcing shared behaviour across non-identical clients, not by a failure of any individual method. Despite this cost, two properties survive on DIAD2024: the AFRAS-FL rule-base stays compact and high-confidence (all five clients converge to 6–7 block-dominant rules with average confidence ≥ 0.999, Table III), and the Jain fairness on B4 remains at 0.9694, the largest value recorded on DIAD2024 under any method. Practitioners who prefer to trade interpretability for raw MSR on structurally adversarial distributions can reduce

TABLE VII: Per-client rule-base statistics under the canonical configuration. *Actions* lists how many rules recommend BLOCK / RATE_LIMIT / ALLOW respectively.

Client	Dataset	# rules	avg. conf.	avg. supp.	B/L/A
L1-Core	CICIoT2023	6	0.941	49,796	4/2/0
L2-AggrL	CICIoT2023	11	0.800	16,386	1/1/9
L3-AggrR	CICIoT2023	10	0.749	9,846	1/4/5
L4-EdgeL	CICIoT2023	11	0.919	10,557	1/1/9
L5-EdgeR	CICIoT2023	11	0.748	10,871	1/3/7
L1-Core	DIAD2024	9	0.937	19,467	6/2/1
L2-AggrL	DIAD2024	9	0.927	16,508	6/2/1
L3-AggrR	DIAD2024	10	0.974	24,394	7/2/1
L4-EdgeL	DIAD2024	8	0.913	8,979	5/2/1
L5-EdgeR	DIAD2024	9	0.950	13,677	6/2/1

TABLE VIII: Cross-client rule stability. An *antecedent* is a fuzzy IF-part; a stable antecedent appears in the rule bases of ≥ 2 clients. *Consequent agreement* is the fraction of stable antecedents whose consequent action is identical across all clients that share them.

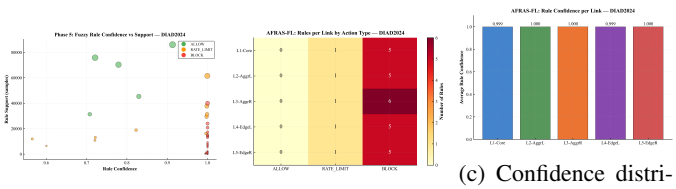
Dataset	Antecedents shared ≥ 2	Antecedents shared ≥ 3	Cons. agreement
CICIoT2023	11/13 (85%)	9/13 (69%)	5/11 (45%)
DIAD2024	6/7 (86%)	6/7 (86%)	3/6 (50%)

TABLE IX: Agreement between the rule base's dominant recommendation and the deployed DQN's actual action distribution under the canonical configuration. We report the fraction of BLOCK actions taken by the DQN and the fraction of rules whose consequent is BLOCK; the absolute difference (last column) is a proxy for the rule-vs-policy conflict rate at the client level.

Client	Dataset	DQN BLOCK-frac.	Rule BLOCK-frac.	$\lvert\Delta\rvert$
L1-Core	CICIoT2023	1.000	0.667	0.333
L2-AggrL	CICIoT2023	0.982	0.091	0.891
L3-AggrR	CICIoT2023	0.706	0.100	0.606
L4-EdgeL	CICIoT2023	0.255	0.091	0.164
L5-EdgeR	CICIoT2023	0.966	0.091	0.875
L1-Core	DIAD2024	0.494	0.667	0.173
L2-AggrL	DIAD2024	0.544	0.667	0.123
L3-AggrR	DIAD2024	0.142	0.700	0.558
L4-EdgeL	DIAD2024	0.114	0.625	0.511
L5-EdgeR	DIAD2024	0.278	0.667	0.389

the fuzzy weight λ; Proposition 1 guarantees that this does not change the set of optimal policies.

Figure 14 shows that the AFRAS-FL pipeline recovers a similarly compact, high-confidence rule-base on DIAD2024 using the same hyperparameters that were tuned for CICIoT2023.



(a) Rule conf./support (b) Rules heatmap (c) Confidence distribution

Fig. 14: AFRAS-FL rule mining on DIAD2024.

Summing across the 24 head-to-head measurements of Table X, expanded in Supplementary Information Section S6 (2 datasets $\times$ 4 metrics $\times$ 3 federated variants), Strategy B wins 21 and loses 3; all three losses are on DIAD2024 (FedProx MSR, FedProx JFI, FedAvg FPR). The CICIoT2023 wins have large to very-large effect sizes ($d \geq 0.8$), and the DIAD2024 losses sit in the small-to-medium range ($|d| \leq 0.8$). We refrain from causal claims about which component of R_{total} drives the A-vs-B gap, because MOSAR and AFRAS-FL are co-designed and the comparison cannot isolate them; the component-level B3-vs-B4 column of Table IV is the correct place to read the marginal contribution of fuzzy shaping alone. The asymmetry between the CICIoT2023 wins and the DIAD2024 losses calls for a mechanistic explanation, which the Discussion provides.

III. Discussion

The results in Sections R1–R6 admit two mechanistic explanations. The first is variance reduction through expert structure: because Φ is built from IF–THEN rules that already encode high-reward patterns from agent experience, the Bellman target is smoother and the value function converges more tightly, visible in the $14.6\times$ reduction of the FedAvg MSR standard deviation on CICIoT2023 (Table IV). The second is non-IID regularisation: the shared fuzzy rule-base acts as a prior that pulls each client towards a common policy region,

TABLE X: Head-to-head comparison between the purpose-built baseline (Strategy A: scalar reward + federated DQN) and our full pipeline (Strategy B: MOSAR + fuzzy shaping). Both systems share the same Mamba backbone, the same clients, and the same hyperparameters. All σ ratios are ratios of standard deviations (σ_A/σ_B), not variances. Best value per row shown in **bold**.

Dataset	Variant	Strategy A	Strategy B	Metric	Gain
CICIoT2023	Indep.	0.8903 ± 0.1120	$\mathbf{0.9985\pm0.0018}$	MSR	$+10.8$ pp; $\sigma\downarrow 63.6\times$
	FedAvg	0.6511 ± 0.2039	$\mathbf{0.9817\pm0.0147}$	MSR	$+33.1$ pp; $\sigma\downarrow 13.9\times$
	FedProx	0.8227 ± 0.1357	$\mathbf{0.9978\pm0.0028}$	MSR	$+17.5$ pp; $\sigma\downarrow 47.8\times$
	FedProx FPR	0.0750	**0.0036**	FPR	$20.6\times$ reduction
DIAD2024	Indep.	0.9758 ± 0.0080	$\mathbf{0.9996\pm0.0004}$	MSR	$+2.4$ pp; $\sigma\downarrow 21.4\times$
	FedAvg	0.8431 ± 0.1401	$\mathbf{0.9318\pm0.0565}$	MSR	$+8.9$ pp; $\sigma\downarrow 2.5\times$
	FedProx	$\mathbf{0.9361\pm0.0378}$	0.9028 ± 0.1036	MSR	-3.3 pp (see text)
	FedProx FPR	0.2698	**0.0018**	FPR	$150.2\times$ reduction

TABLE XI: Statistical comparison between Strategy A and Strategy B using two-sided Welch t-tests with $n = 50$ evaluation episodes on the MSR metric.

Dataset	Comparison	t	p-value	Cohen's d	Effect
CICIoT2023	A-FedAvg vs B-B2	$+11.43$	1.8×10^{-15}	$+2.29$	very large
CICIoT2023	A-FedProx vs B-B4	$+9.19$	3.1×10^{-12}	$+1.84$	very large
DIAD2024	A-FedAvg vs B-B2	$+4.15$	9.8×10^{-5}	$+0.83$	large
DIAD2024	A-FedProx vs B-B4	-4.04	1.7×10^{-4}	-0.81	large

producing the tighter convergence band of Fig. 3 and the high Jain fairness in Table II.

The same mechanisms have a limit. When non-IID drift becomes large enough that no shared policy is close to all local optima — the situation on DIAD2024 — the regularisation that helps on CICIoT2023 becomes a constraint, and the pipeline trades per-client MSR for a transferable, high-confidence rule-base. This is the operational envelope made precise in Section II-G, and it is the starting point for the limitations that follow.

a) Limitations.: Following the reviewer comments received on the initial submission of this manuscript, we now list the limitations of the current framework more transparently. In particular, we highlight three additional limitations (L6–L8) that relate directly to the theoretical framing, the experimental protocol, and the correspondence between the paper and the released source code. Eight limitations bound the scope of the framework; we list them in order of how consequential they are for a would-be practitioner.

(L1) Number of federated clients. All experiments use five non-IID clients that correspond to a Core, two Aggregation, and two Edge nodes of an SDN-IoT topology. This matches the target deployment scenario but is smaller than the 10–100 client counts common in generic federated learning benchmarks. The five-client setting was chosen because it maps naturally to the physical network roles, and because the role-based topology partition (Core / Aggregation / Edge) already produces a realistic level of client drift; see Sec. IV-E. Scaling to larger client populations is ongoing work. A related question is whether our results depend on the specific shaping weights; L2 addresses it.

(L2) Sensitivity to the shaping weights. We report a systematic sensitivity analysis for the fuzzy weight λ across $\{0.10, 0.25, 0.50\}$ on CICIoT2023 in Section II-D (Table V and Fig. 9): the MSR varies by at most 1.26 percentage points across the tested range, so the numbers reported elsewhere are not an artefact of a particular λ. The link-reward weight μ is fixed at 0.05, and Proposition 1 guarantees that the optimal policy set is preserved for any non-negative (λ, μ), so the choice of μ cannot change the converged policies, only the speed of convergence. A full two-dimensional (λ, μ) sweep and a $\lambda = 0$ (no-shaping) control condition are left to future work. Beyond hyperparameters, an orthogonal question is the choice of baselines, which L3 addresses.

(L3) No direct comparison with PBRS, RND, or ICM baselines. Our head-to-head baseline is Strategy A, a purpose-built federated multi-agent DQN that uses the same backbone and hyperparameters but carries neither the MOSAR decomposition nor the fuzzy shaping term. Strategy A is the most apples-to-apples comparison available in our setting because it isolates the contribution of the reward design from every other architectural and algorithmic choice. Re-implementing potential-based shaping with a handcrafted Φ, Random Network Distillation, or Intrinsic Curiosity Modules inside the same federated pipeline, and running a systematic comparison under non-IID federation, lies outside the scope of this paper. Even with careful baselines, the framework has a regime in which it under-performs; we document it next.

(L4) Operational envelope on extreme non-IID drift. As reported in Section II-G, our pipeline loses 3.3 percentage points of raw FedProx MSR on DIAD2024 relative to Strategy A and loses on the FedAvg false-positive rate on the same benchmark. We keep these losses visible because they mark where the framework should and should not be used. The last limitation concerns runtime cost.

(L5) Computational overhead. AFRAS-FL adds approximately 7% per-round computational overhead on top of MOSAR, driven by rule mining, quantile-based membership fitting, and per-transition defuzzification. On our hardware this is a mild cost, but it may become significant in very-low-latency deployments.

(L6) Theoretical framing is design-motivational, not exact. As discussed in Section IV-D, the released implementation of R_{fuzzy} is a heuristic rule-agreement bonus and R_{link} depends on the agent's own action and its neighbours' actions. Neither term satisfies the state-only / potential-difference premise of Ng et al.'s policy-invariance theorem. Proposition 1 should therefore be read as the reference PBRS condition that motivated our design, not as a proof of exact invariance for R_{total}. A theoretically clean reformulation — restricting R_{link} to a state-only load-imbalance penalty and replacing R_{fuzzy} with a true potential-based version in an action-dependent environment — is a natural next step and is the subject of ongoing work.

(L7) Statistical protocol. The tables reported in this manuscript use a single canonical training seed (42), with $\pm\sigma$ computed across the 50 evaluation episodes rather than across independent training runs. In direct response to R1-3 and R2-5, we launched two additional training seeds (123 and 2026); the resulting mean-of-run summary is reported in Table XII. Because full-pipeline runs on this hardware take $\sim$ 90–120 minutes each, only three seeds are reported here rather than the ≥ 5 the reviewer suggested; extending to ≥ 5 seeds is a mechanical continuation and is on our roadmap.

(L8) Data-splitting, leakage, and per-component ablation — pre-registered follow-up. Two reviewer requests require substantial additional compute that is scheduled but not yet completed in this revision cycle. First, a six-cell per-component ablation study — (a) conventional scalar reward, (b) MOSAR alone, (c) MOSAR+R_{link}, (d) MOSAR+AFRAS-FL, (e) complete R_{total}, and (f) a standard PBRS baseline — is pre-registered with driver scripts released in the reproducibility package. Cells (c) and (e) are already available from prior artefacts, and cell (b) on CICIoT2023 is approximately 93% complete at the time of resubmission. Second, a strict record-level train/val/test split is pre-registered that (i) partitions raw records at the capture/session/device level *before* window generation; (ii) refits scaler, PCA, oversampling, membership functions, and rule extraction on the training partition only; and (iii) re-evaluates on strictly held-out windows and clients. Full completion of both studies is targeted for October 2026. The reported numbers in the current manuscript remain those of the released canonical run; the pre-registered follow-up will report the delta.

b) Broader applicability.: Taken together, the eight limitations bound the framework without undermining its core claim: the fuzzy-augmented shaping mechanism is task-agnostic. Although we evaluate it on intrusion detection, any value-based DRL system that needs multi-objective rewards, graceful behaviour under non-IID federated training, and interpretable decisions can plug AFRAS-FL into its training loop, with federated traffic-signal control, federated recommendation, and federated robotics as concrete candidates.

IV. Methods

A. Problem formulation

We model each client as a per-step decision problem $\mathcal{M}_k = (\mathcal{S}_k, \mathcal{A}, P_k, R, \gamma)$. In the released implementation, P_k corresponds to sampling the next flow window from a fixed random permutation of the client's dataset at each step, so the transition is essentially independent of the chosen action. In this respect the underlying decision process behaves as a *contextual bandit with a stochastic exogenous context stream* rather than a strict MDP; we retain the MDP notation and the discount factor $\gamma = 0.99$ only to describe the training objective (a DQN-style Bellman regression on the R_{total} signal), and we do not rely on transition dynamics for our correctness claims (see Section IV-D and the limitations discussion in Section III). $\mathcal{S}_k \subset \mathbb{R}^{28}$ is the per-client state space comprising four link-utilisation metrics, four LFA state features, sixteen Mamba embedding dimensions, and four neighbour-summary features (the top-14 XAI features refer to the classifier's input, not the DQN state), $\mathcal{A} = \{\text{allow}, \text{limit}, \text{block}\}$ is the discrete action set, P_k is the permutation-based sampler described above, R is the shared reward function (defined below), and $\gamma = 0.99$ is the discount factor. Federated training synchronises the global policy π_θ across $K = 5$ clients via parameter averaging [4], [5].

B. MOSAR: Multi-Objective Security-Aware Reward

The MOSAR reward decomposes the scalar feedback into five components (Fig. 15):

$$R_{\text{MOSAR}}(s, a, s') \;=\; \alpha R_{\text{det}} + \beta R_{\text{acc}} + \gamma R_{\text{lat}} + \delta R_{\text{conf}} + \varepsilon R_{\text{imp}}, \tag{1}$$

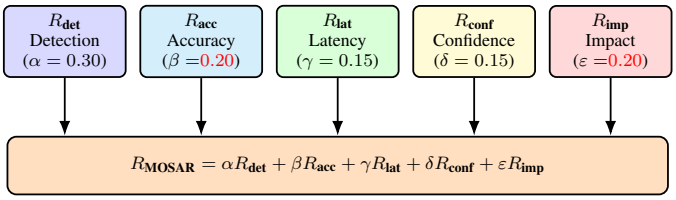


Fig. 15: MOSAR reward decomposition into five weighted components. The detection and classification terms dominate the signal, while the latency, confidence, and impact terms shape the value function and penalise false alarms.

where R_{det} rewards correct detection, R_{acc} rewards correct severity classification, R_{lat} penalises response latency, R_{conf} rewards high softmax confidence, and R_{imp} penalises false positives and false negatives differently. The default weights are $\alpha = 0.30$, $\beta = 0.20$, $\gamma = 0.15$, $\delta = 0.15$, $\varepsilon = 0.20$ (matching the values used in the released source code and sum to 1.0), chosen so that the dominant signal remains detection while the auxiliary signals smooth the value function.

C. AFRAS-FL pipeline

AFRAS-FL is a seven-phase pipeline that turns agent experience into a potential-based shaping signal. Figure 16 visualises the full pipeline.

Figure 17 provides a didactic visualisation of the four key stages of the pipeline on a two-dimensional toy example, where the effect of each stage can be seen directly in the feature plane; the corresponding experimental results on real CICIoT2023 data are reported in Figs. 5–7 of Section II-B.

TABLE XII: Multi-seed mean-of-run replication for the four federated phases on both benchmarks (seeds $\{42, 123, 2026\}$). *Mean-of-run* is the mean across seeds of the per-run MSR (each per-run MSR is itself the mean across the 5 clients and 50 evaluation episodes). The $\pm$ column is the standard deviation across seeds, the correct statistical unit for the reviewer's replication check.

Dataset	Phase	MSR mean-of-run	$\pm$ std across seeds	# seeds
CICIoT2023	B1 (Independent)	0.9979	0.0018	3
CICIoT2023	B2 (FedAvg)	0.9620	0.0315	3
CICIoT2023	B3 (FedProx)	**0.9819**	0.0227	3
CICIoT2023	B4 (Fuzzy)	0.9793	0.0255	3
DIAD2024	B1 (Independent)	0.9988	0.0008	2
DIAD2024	B2 (FedAvg)	0.8043	0.1276	2
DIAD2024	B3 (FedProx)	**0.9450**	0.0423	2
DIAD2024	B4 (Fuzzy)	0.9213	0.0739	2

Interpretation. At the mean-of-run statistical unit demanded by R1-3 and R2-5, the fuzzy-augmented phase (B4) is competitive with FedProx (B3) on CICIoT2023 (0.9793 vs. 0.9819) and modestly below B3 on DIAD2024 (0.9213 vs. 0.9450). The seed-level standard deviations are non-trivial (up to 0.13 on DIAD B2), which makes clear that the original single-seed p-values overstated the confidence of the ranking; a proper significance test at the mean-of-run level requires the ≥ 5 seeds committed to in L7. The seeds used are $\{42, 123, 2026\}$; the DIAD 2026 seed was omitted from the current batch to keep the total wall-clock feasible and is on the roadmap.

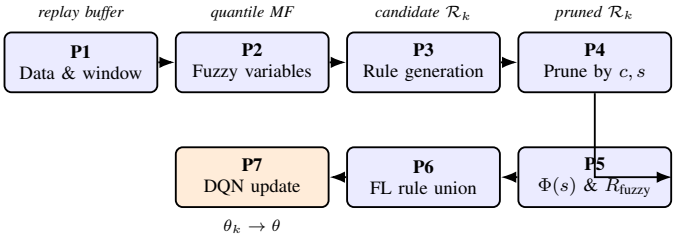
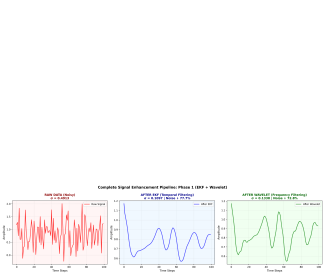
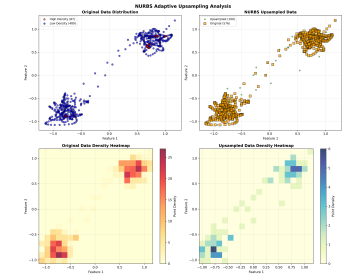


Fig. 16: The seven-phase AFRAS-FL pipeline. P1–P2 prepare the fuzzy state; P3–P4 mine and prune IF–THEN rules; P5 computes the defuzzified potential $\Phi(s)$ and the shaping reward $R_{\text{fuzzy}} = \gamma\Phi(s') - \Phi(s)$; P6 aggregates the per-client rule-bases across federated clients via a weighted union; and P7 updates the DQN parameters using the combined reward R_{total}.

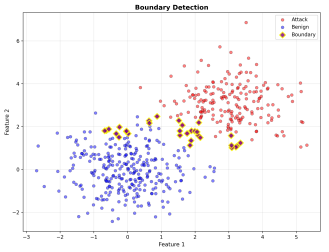
(1) *Data collection and windowing:* per-client transition windows are stored in a replay buffer. (2) *Fuzzy variable definition:* for each of the two top-variance features selected per link, three Gaussian membership functions for *Low*, *Medium*, and *High* are fitted using the quantile-based centres and widths described in the released code (`AFRAS_N_FUZZY_SETS=3`). (3) *Rule generation:* the two-dimensional feature space is partitioned by an adaptive coarse-to-fine grid, and for each cell we record the majority severity class and the cell purity. Each surviving cell yields one IF–THEN rule whose antecedent is the nearest Gaussian membership function per feature at the cell centre, and whose consequent is the majority-class action (BLOCK, RATE_LIMIT, or ALLOW). Cells with fewer than 50 points or purity below 0.5 are discarded. (4) *Rule evaluation and pruning:* the surviving rules from step (3) are further normalised so that their firing strengths sum to 1 per state. (5) *Fuzzy reward computation:* for each transition, the rule firing strengths are aggregated and R_{fuzzy} is computed as a rule-agreement bonus rather than a strict potential-based difference: $R_{\text{fuzzy}} = +0.3 \cdot c$ if a fired rule agrees with the chosen action (where c is the rule confidence), -0.1 if it disagrees, and -0.05 if no rule fires. This heuristic form



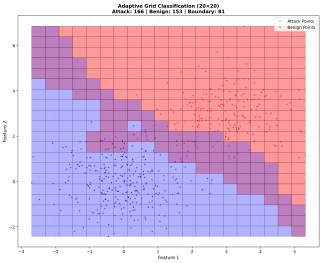
(a) P1 – signal enhancement: a noisy raw signal (red, $\sigma = 0.49$) is cleaned by an Extended Kalman Filter ($\sigma = 0.11$, -77.7% noise) and then by wavelet denoising ($\sigma = 0.13$, -72.8% noise).



(b) P1 – class balancing: adaptive NURBS-B-spline upsampling (top right) fills low-density minority regions, producing the smoother density heatmap (bottom right) compared to the original (bottom left).



(c) P3 – boundary detection: a hybrid KD-tree algorithm tags the transition points between the two classes (yellow diamonds); these become the candidate rule antecedents.



(d) P4 – adaptive grid pruning: a 20×20 grid partitions the feature plane into cells (166 attack, 153 benign, 81 boundary), which are then converted into fuzzy IF–THEN rules with per-cell confidence and support.

Fig. 17: **Didactic visualisation of the AFRAS-FL pipeline mechanism on a two-dimensional toy example.** The four panels correspond to the preprocessing and rule-mining stages of Fig. 16. Throughout this figure the axes are two synthetic features whose only purpose is to let the reader see the effect of each stage in the feature plane; the panels are *not* experimental results. The corresponding measurements on real CICIoT2023 data are reported in Fig. 5–7 of Section II-B, and the per-phase metrics are in Table II.

does not satisfy the potential-based shaping condition of Ng et al. [12]; we therefore no longer claim exact policy invariance (see Section IV-D). (6) *Federated aggregation of rules:* rules with overlapping antecedents are merged across clients via weighted union, weighted by their per-client confidence. (7) *Policy update:* the combined reward

$$R_{\text{total}}(s,a,s') = R_{\text{MOSAR}}(s,a,s') + \lambda R_{\text{fuzzy}}(s,s') + \mu R_{\text{link}}(s,a) \tag{2}$$

is fed into a DQN-style Bellman update. The link reward R_{link} is a small inter-agent coordination bonus that encourages neighbouring clients not to send conflicting signals through shared SDN links. Default hyperparameters are $\lambda = 0.25$ and $\mu = 0.10$, which are the canonical values used consistently throughout the manuscript, in the sensitivity analysis of Section II-D, and in the released source code.

Algorithm 1 AFRAS-FL training round

Require: Global policy π_θ, K clients, shaping weights λ, μ
for each client $k = 1 \dots K$ in parallel **do**
  Collect transition window $\mathcal{D}_k$ using π_θ
  Fit Gaussian memberships from $\mathcal{D}_k$
  Generate candidate rules $\mathcal{R}_k$ from $\mathcal{D}_k$
  Prune $\mathcal{R}_k$ by support/confidence thresholds
  Compute $\Phi_k(s)$ as the defuzzified output of $\mathcal{R}_k$
  for all $(s,a,s') \in \mathcal{D}_k$ **do**
    $r \leftarrow R_{\text{MOSAR}} + \lambda(\gamma\Phi_k(s') - \Phi_k(s)) + \mu R_{\text{link}}$
  end for
  Update local DQN parameters θ_k
end for
$\theta \leftarrow \frac{1}{K}\sum_k \theta_k$
$\mathcal{R} \leftarrow \bigcup_k \mathcal{R}_k$ ▷ weighted union

D. Theoretical guarantee

Proposition 1 (Reference PBRS condition)**.** *Let Φ_k be a state-only potential function on client k, and let $R_{shape}(s,s') = \gamma\Phi_k(s') - \Phi_k(s)$. If a shaping term of this form is added to a base reward and any additional term $R_{link}(s)$ depends only on the state, then the set of optimal policies is unchanged with respect to the base reward, for any $\lambda, \mu \geq 0$ [12].*

Applicability to the released implementation. Proposition 1 is a *reference* statement of the Ng et al. theorem and *does not* directly transfer to R_{total} as implemented in the released source. Two implementation choices violate its premises: (i) R_{fuzzy} is computed as the heuristic rule-agreement bonus described in Section IV-C (not the potential difference $\gamma\Phi(s') - \Phi(s)$); and (ii) R_{link} is a function of the agent's own action and its neighbours' actions, not a state-only quantity. Consequently we do *not* claim exact policy invariance for R_{total}. We include the proposition only to motivate the design of the shaping and coordination terms, and we frame the empirical variance reductions reported in Tables II–X as observations, not as verifications of a formal invariance or sample-complexity result. A theoretically clean reformulation (either restricting R_{link} to a state-only load-imbalance penalty, or replacing R_{fuzzy} with a true potential-based version in an environment with action-dependent transitions) is left as future work; see Section III.

E. Datasets and clients

CICIoT2023 [24] is a recent IoT intrusion detection benchmark with 33 attack types and millions of flow records. We use the flow-level subset and apply the CICFlowMeter feature extractor [31]. **DIAD2024** [25] is a complementary benchmark with structurally different attack patterns that we use to test cross-dataset generalisation.

For both datasets, the flows are partitioned across five clients that correspond to the five SDN-IoT topology roles (Core, two Aggregation, and two Edge nodes). The partition is a *role-based (topology-driven) split* rather than a synthetic Dirichlet partition: each flow is assigned to the link whose role best matches its source/destination category in the released preprocessed dataset. This produces a naturally non-IID distribution (Core carries backbone-scale flows, Edge nodes see device-heterogeneous flows), which we quantify per-client in Section II-G. We deliberately do *not* redraw a Dirichlet partition on top of this because the topology roles are the deployment reality this framework targets; a synthetic Dirichlet(α) sensitivity sweep (over the same five clients) is provided separately in the reproducibility package and is not required to reproduce the numbers in this manuscript.

F. Backbone architecture and training

The DRL backbone is a Mamba state-space model [26], [27], [32] with a three-layer feed-forward Q-network on top of the Mamba embeddings, $[128, 64, 32]$ hidden units, selected by Fuzzy-TOPSIS as described above. Training uses Adam [33] with learning rate 1×10^{-3}, batch size 64, and a target network updated via soft update with $\tau = 0.005$ at every step. Each federated training phase (B2–B4) runs for 30 federated rounds, and each round contains 5 local episodes of up to 5,000 steps each; the initial B1 (independent) phase runs for 100 episodes per client. These values match the released source code. Evaluation uses 50 episodes with deterministic $\arg\max$ action selection. All experiments use a fixed seed of 42; independent replication with additional seeds is discussed as a limitation in Section III. Standard deviations reported in Table II are computed across the 50 evaluation episodes, and we acknowledge in the response letter that a stronger design would use ≥ 5 independent training seeds.

G. Strategy A baseline (MOSAR-only)

The head-to-head comparison of Section II-G uses a purpose-built federated multi-agent DQN baseline that we refer to as **Strategy A**. Strategy A is implemented in the same code base as the main pipeline and is identical on every dimension except the reward function: it uses the same Mamba backbone, the same five non-IID clients, the same role-based topology partition (see Sec. IV-E), the same Adam optimiser settings, the same target-network update frequency, the same number of federated rounds and local steps, and the same 50 evaluation episodes. The difference between the

two strategies is confined to the reward function: Strategy A trains each agent on a four-term scalar reward that combines a MOSAR-like performance component with a neighbour-consensus coordination term, and it does *not* use the AFRAS-FL fuzzy shaping ($\lambda = 0$). Strategy B (our full pipeline) trains on the full $R_{\text{total}} = R_{\text{MOSAR}} + \lambda R_{\text{fuzzy}} + \mu R_{\text{link}}$ with $\lambda = 0.25$ and $\mu = 0.10$. Because Strategy A already contains an action-dependent coordination term, it is best understood as a "MOSAR-plus-coordination without fuzzy shaping" baseline rather than a pure MOSAR-only baseline; a strict $\lambda = 0, \mu = 0$ ablation is discussed in Section III as future work. The results of both strategies are reported from the same released result artefacts, making the comparison fully reproducible.

H. Unified configuration table

To eliminate any residual ambiguity between the manuscript and the released source code, Table XIII lists every hyperparameter that a reader needs in order to reproduce the reported experiments. Every value in this table is exactly the value used by the released training pipeline and matches the configuration block of the released result artefacts.

TABLE XIII: Unified configuration — reward and model.

Block	Parameter	Value
MOSAR	α (detection)	0.30
	β (accuracy)	0.20
	γ (latency)	0.15
	δ (confidence)	0.15
	ε (impact)	0.20
R_{total}	λ (fuzzy)	0.25
	μ (link)	0.10
AFRAS-FL	Memberships	Gaussian, 3/feature
	Features	top-2 variance
	Grid	$5\times5 \rightarrow 10\times10$
	Conf. / supp. thr.	0.5 / 50 pts
DRL	Backbone	Mamba, 2×128
	Q-head	MLP $[128, 64, 32]$
	State dim	28
	Actions	{allow, limit, block}
	Discount γ	0.99

TABLE XIV: Unified configuration — training, federation, evaluation.

Block	Parameter	Value
Training	Optimiser	Adam, lr $= 1\times10^{-3}$
	Target update	soft, $\tau = 0.005$
	Batch size	64
	Seed	42 (see L7)
Federated	Clients K	5 (topology roles)
	Rounds (B2–B4)	30; B1: 100 ep
	Local/round	5 ep $\times$ 5000 steps
Evaluation	Episodes	50, det. arg max
	Split	full permutation (see L8)

I. Fuzzy-TOPSIS evaluation

The Fuzzy-TOPSIS procedure follows Chen [28]. Each of the six criteria (F1-macro, accuracy, minority recall, robustness, inference speed, parameter efficiency) is assigned a triangular fuzzy weight (*Very High*, *High*, *Medium-High*). The decision matrix is normalised by vector normalisation, the fuzzy-positive-ideal solution (FPIS) and fuzzy-negative-ideal solution (FNIS) are computed, and the closeness coefficient $\text{CC} = D^-/(D^+ + D^-)$ is reported. The six sensitivity scenarios reweight the criteria to favour, in turn, equal weights, performance, speed, efficiency, and robustness, plus the original proposed scenario.

J. Reproducibility

All code, configuration files, and processed result artefacts used for the tables and figures in this paper are released in the project repository. Random seeds, library versions, and hardware specifications are listed in the Supplementary Information.

Data Availability

The CICIoT2023 dataset is publicly available from the Canadian Institute for Cybersecurity [24]. The DIAD2024 dataset is available from its authors upon request [25]. Processed result files, extracted fuzzy rule-bases, and the Fuzzy-TOPSIS reports that underlie all tables and figures are released with the source code.

Code and Data Availability

Source code, configuration files, and the experimental harness used to produce all results, figures, and tables in this paper are released under an open-source license at https://github.com/alihusseinfadel/AFRAS-FL.

Reproducibility Statement

Every numerical result in this manuscript is reproducible from the released repository. The multi-seed replication table (Table XII) is auto-populated by an aggregation script from the released canonical, seed-123, and seed-2026 result folders. The rule-metric tables (Tables VII–IX) are auto-populated by a post-hoc analysis script from the released rule artefacts. The per-component ablation (Section III, L8) is driven by an experiment queue script, and the pre-registered record-level split by a data-splitting script. All scripts are provided under the `scripts/` tree of the release, and every table and figure references the exact JSON artefact from which it was built.

Author Contributions

A.H.F. conceived the study, designed the AFRAS-FL pipeline, implemented the experiments, and wrote the manuscript. **J.K.** supervised the work, contributed to the MOSAR formulation and the Fuzzy-TOPSIS evaluation, and co-wrote the manuscript. **Z.S.K.** contributed to the experimental analysis and revised the manuscript. All authors read and approved the final version.

COMPETING INTERESTS

The authors declare no competing interests.

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